

README

The dataset contains image patches (`*.jpg`) with oil slicks, look-alikes, or other remarkable phenomena. These image patches are categorised into *oil* and *no-oil* sets for image patches with and without oil slicks inside.

Note

For the image patches containing oil slick(s) (i.e. in the *oil* set), the locations of the oil slicks are recorded in corresponding `*.xml` files, following the PASCAL VOC XML annotation format commonly used by object detection algorithms.

Inside each of these two sets, image patches with and without land coverage are categorised into "coast" and "water" subsets. The basic idea is to allow users to understand how their models perform in different subsets.

The structure of different subsets can be understood like this:

```
oil/  
  coast/  
  water/  
no_oil/  
  coast/  
  water/
```

However, the submitted dataset avoids the complicated folder structure; the information mentioned above is recorded in the image patch name, using an identical tag following the naming convention:

```
00-0000[-xx-xxxxxx]
```

`00`: 2-character subset tag referring to different subsets

`oc`: **oil/coast**

`ow`: **oil/water**

`nc`: **no_oil/coast**

`nw`: **no_oil/water**

`0000`: image serial number, each subset counts from `0001`

`xx`: object index (*oil set*) / cluster index (*no-oil set*)

object index: indexing the objects in each image patch, counting from `01`

cluster index: image patches inside *no-oil* set are classified by the K-Means clustering

`no_oil/coast` → clustered into 5 categories, the cluster index: `00`, `01`, `02`, `03`, `04`

`no_oil/water` → clustered into 12 categories, the cluster index: `00`, `01`, `02`, ..., `11`

`xxxxxx`: object serial number (*oil set*) / cluster patch serial number (*no-oil set*)

object serial number: indexing the object in each subset, starting from `000001`

cluster patch serial number: indexing the image patches in each class, starting from `000001`

For the *oil* set (with tags starting with `oc` & `ow`), as the object index and object serial number (`xx-xxxxxx`) refer to an object, not an image, the image patch name is only using the subset tag and image serial number (`oo-0000`). However, the tag is used as a reference index in the data table (`data_table.xlsx`), which records all relevant information about each image patch and each oil object (explained in page 3).

For example, the tags in the first few rows of the `oc` tab in `data_table.xlsx` and their explanations are shown in the following table:

tag	corresponding image	notes
<code>oc-0001-01-000001</code>	<code>oc-0001.jpg</code>	oil #1 (01) in image <code>oc-0001.jpg</code> ; the <code>000001</code> object in the subset <code>oc</code>
<code>oc-0001-02-000002</code>	<code>oc-0001.jpg</code>	oil #2 (02) in image <code>oc-0001.jpg</code> ; the <code>000002</code> object in the subset <code>oc</code>
<code>oc-0001-03-000003</code>	<code>oc-0001.jpg</code>	oil #3 (03) in image <code>oc-0001.jpg</code> ; the <code>000003</code> object in the subset <code>oc</code>
<code>oc-0001-04-000004</code>	<code>oc-0001.jpg</code>	oil #4 (04) in image <code>oc-0001.jpg</code> ; the <code>000004</code> object in the subset <code>oc</code>
<code>oc-0001-05-000005</code>	<code>oc-0001.jpg</code>	oil #5 (05) in image <code>oc-0001.jpg</code> ; the <code>000005</code> object in the subset <code>oc</code>
<code>oc-0002-01-000006</code>	<code>oc-0002.jpg</code>	oil #1 (01) in image <code>oc-0002.jpg</code> ; the <code>000006</code> object in the subset <code>oc</code>

Tip

Image patches are separated into *oil* and *no-oil* sets depending on whether they contain oil slick(s) or not. So users can expect the following files for oil and no-oil sets:

- *oil* set: `oo-000000.jpg` & `oo-000000.xml`
- *no-oil* set: `oo-000000-xx-xxxxxx.jpg`

Data table (`data_table.xlsx`) includes the following fields:

Fields	Explanations
<code>tag</code>	as explained above (<code>00-0000-xx-xxxxxx</code>)
<code>patch_name</code>	the original patch name, which was assigned while generating the image patches, it follows the naming convention: <code>MM_YYYYMMDD_HHMMSS_HHMMSS_pp_i</code> * <code>MM</code> : satellite mission, <code>S1</code> for Sentinel-1 * <code>YYYYMMDD</code> : date of the product * <code>HHMMSS</code> : start and stop time of the product * <code>pp</code> : polarisation mode * <code>i</code> : a series of numbers assigned while generating the image patches
<code>start_time</code> , <code>end_time</code>	start and stop date time of the product
<code>Sentinel_ID</code>	Sentinel-1 product ID for the image patch (as more than one Sentinel-1 scenes were processed together, there might be more than one ID corresponding to one image patch)
<code>patch_width</code> , <code>patch_height</code>	image patch dimension
<code>patch_ul_lon</code> , <code>patch_ul_lat</code> , <code>patch_ur_lon</code> , <code>patch_ur_lat</code> , <code>patch_br_lon</code> , <code>patch_br_lat</code> , <code>patch_bl_lon</code> , <code>patch_bl_lat</code>	the geolocation of the corner coordinates of the image patch in the WGS84 geocentric coordinate system
<code>obj_ul_lon</code> , <code>obj_ul_lat</code> , <code>obj_ur_lon</code> , <code>obj_ur_lat</code> , <code>obj_br_lon</code> , <code>obj_br_lat</code> , <code>obj_bl_lon</code> , <code>obj_bl_lat</code>	the geolocation of the corner coordinates of the object in the WGS84 geocentric coordinate system (only for <i>oil</i> set)
<code>obj_patchloc_xmin</code> , <code>obj_patchloc_ymin</code> , <code>obj_patchloc_xmax</code> , <code>obj_patchloc_ymax</code>	image coordinates of the object referring to its corresponding image patch (only for <i>oil</i> set)
<code>label_size</code>	the bounding box size in pixel (only for <i>oil</i> set)

The original patch name (`patch_name`) was assigned while generating the image patches; it carries information about the date and time of the acquisitions. Users can also rename the files to the patch name as they wish. However, it shall be noted that as *oil* and *no-oil* sets were created separately, there might be duplicated patch names. If users wish to rename the files, they are suggested to create folders for different subsets. A suggested folder structure is shown as follows:

```
oil/  
  coast/  
  water/  
no_oil/  
  coast/  
    c00/  
    c01/  
    ..  
    c04/  
  water/  
    c00/  
    c01/  
    ...  
    c11/
```

For detailed explanation on the dataset, please refer to the [data descriptor](#).